

S27 Research Topic Ideation

Machine Learning × Topological Data Analysis × Nick Morihisa's background × the market

Prepared 2026-10-02 from the AMAT 583 / 584 / 592 materials in Google Drive, the AMAT 581 catalog description, resumes and side projects, and public work by Scott Evans (grid optimization, energy-aware ML) and Barbara Giunti (applied topology, multiparameter persistence). Market-size figures are 2026 analyst estimates that vary by firm; ranges are given and sources appear on the last page. The ★ marks the two topics judged the strongest fit for both professors.

Inputs reviewed

Source	What it contained	How it shaped the ideas
AMAT 583 — Topological Data Analysis I (Drive: AMAT 583/, Module-01 ... Module-20, TDA I HW Problems.pdf)	Graph theory, simplicial complexes, Vietoris–Rips and Čech complexes, nerve theorem, chain complexes and chain maps, simplicial homology and simplicial maps, persistent homology, continuity and deformations (homotopy invariance).	Core PH vocabulary: filtrations, Betti numbers, barcodes.
AMAT 584 — Topological Data Analysis II (Drive: L0-intro.pdf ... L13-sec.3.2.pdf)	Ch.1 persistent homology as an ML feature source (coordinate invariance, 1-Lipschitz stability Hausdorff→bottleneck, custom filtrations, nonlinearity), sublevel-set filtrations, barcode metrics and stability; Mapper and its applications (cancer-subgroup example, feature selection); Ch.2 topology and cellular (CW) complexes, cellular homology, winding numbers; Ch.3 classical and discrete Morse theory (Forman).	Stability theorem as a robustness argument; Mapper for archetype discovery; multi-filtration thinking.
AMAT 592 — Machine Learning (Drive: * notes.pdf, Aug 2026; AMAT 592 Final Project Report 2026 NM)	OLS/ridge/MLE, cross-validation, Bayes and naive Bayes classifiers, logistic regression, perceptron, SVM and kernel trick, KDE, k-means, neural networks, SGD, learning theory (ERM, Hoeffding, uniform deviation bounds), performance metrics, under/over-fitting. Final project: Olist delivery-time regression — stage decomposition, variance-decomposition R² ceiling ≈ 0.35 , date-block splits vs. chronological drift, quantile (pinball, τ=0.90) delivery promise beating the incumbent by 2.5 days, loss-function vs. model-family variance.	A reusable methodology: "measure how predictable a target is before modeling it," plus calibrated quantile promises.
AMAT 581 — Nonparametric Statistics (no files in Drive yet; UAlbany graduate bulletin)	Statistical tools for modern data-science problems with ML emphasis: boosting, online learning, large-scale data analysis, hands-on software.	Online learning / boosting / conformal-style distribution-free methods for streaming and shifting data.
Background (resumes 02.26 / 05.26 / 08.26, Energy Learning Path.xlsx, NM_Mat425_VPPs_SAA.docx, NYC-Asian-Food-Information-Gaps-Report, Quantum Talent take-home)	Harvard Economics; Walmart eCommerce category buyer / data strategy (\$10M P&L, shipping data); Terra2 solar investment startup; PULP \$200M progressive energy-rate program; NYS DEC emissions and refrigerant (Rule 494) databases; Sen. Fahy legislative team with a multi-model LLM policy-synthesis pipeline; campaign field director / manager; professional poker (solvers, risk limits, 4.5 bb/100); K-8 special-ed math teacher; Murals for Good Chinatown (\$1.3M budget); MAT 425 paper on Sample Average Approximation convergence for VPP dispatch; Energy Learning Path tracks (NYISO LMP/congestion, forecasting, DER equity, BESS TEA).	Domain hooks: energy/grid, marketplace logistics, legislative AI, decision-making under uncertainty, urban/community data, education.
Scott Evans (UAlbany Math & Stats lecturer; Principal Scientist ML, GE Research / GE Vernova)	Computational math, optimization, tensor computation, information theory; "Sustainable Grid through Distributed Data Centers: Spinning AI Demand for Grid Stabilization and Optimization" (2025, IEEE); patent on carbon-aware scheduling of ML jobs across the grid (LSTM demand forecast + dispatch); sparse-tableau optimal power flow; ARPA-E summit demos; current focus you noted: electrical grid layout optimization and energy-load problems.	Grid layout, OPF, forecasting, data-center flexible load.
Barbara Giunti (UAlbany; applied and computational topology)	Multiparameter persistence (efficient two-parameter computation via cohomology; fast queries of fibered barcodes); "Canopies: a generalization of vines and vineyards for parameterized persistence" (with E. Munch, 2026); Galois connections in PH; Grassmannian persistence diagrams; "Comparing Chatbot Performance Enhanced with Persistent Homology" (with N. Raghavaraju, B. Rieck, June 2026); "SuPerPoV: score and evolution of the stratospheric polar vortex via persistent homology" (Jan 2026).	Time-varying persistence (vineyards/canopies), bifiltrations, PH features for NLP, PH for climate time series.

Toolkit, compressed

Course	Tools you can deploy in a research project
583	Build complexes from point clouds and graphs (VR, Čech, alpha), compute H0/H1/H2 and persistence, read barcodes, prove invariance under deformation.
584	Use PH as features (vectorizations), argue robustness via the stability theorem, design custom (sublevel-set) filtrations, run Mapper for subgroup discovery, apply discrete Morse theory to simplify complexes; think in multiple filtration parameters.
592	Fit, validate and compare regression/classification/clustering/NN models; use learning-theory bounds; decompose variance to bound achievable accuracy; design leakage-free and shift-aware splits; quantile and robust losses.
581	Boosting, online learning (no-regret), nonparametric estimation and distribution-free inference for streaming and non-stationary data.

Table A — Machine Learning × background × market (no TDA)

#	Topic	Summary	Problems that exist today	Example research questions	Market size (2026, approx.)	Example employers	Fit
A1	Predictability ceilings and quantile delivery promises for marketplaces	Generalize your 592 result (R^2 ceiling ≈ 0.35 ; 6-day unexplained residual; $\tau=0.90$ promise beats incumbent) into a method marketplaces can use to decide whether to buy better models or better observability.	Promise dates set as liability buffers (Olist over-promised by ~ 12 d); no principled test for "model failure vs. missing data"; chronological drift (strikes, Carnival, Black Friday).	Can a group-mean variance-decomposition estimator give a provably valid upper bound on achievable R^2 for a feature set? Do conformal intervals beat pinball-loss quantiles on on-time rate under chronological shift? What is the marginal value of a carrier-scan event vs. a better model?	Last-mile delivery $\approx$ \$200B; supply-chain analytics $\approx$ \$14B; last-mile software $\approx$ \$3B.	Amazon, Walmart GoLocal, Shopify, Mercado Libre, Olist, Flexport, project44, FourKites, UPS, FedEx, Instacart, DoorDash.	Walmart eCommerce + existing 592 code. Evans (optimization/forecasting).
A2	Load and LMP/congestion forecasting under data-center-driven structural shift	Short-term load and nodal price forecasting (GBM, online learners) where new large loads change the data-generating process; the "NYISO LMP & Congestion Explorer" from your Energy Learning Path, made rigorous.	Utilities name data centers the top load driver; models trained pre-2023 degrade; congestion migrates; interconnection-queue uncertainty.	Which online-learning / boosting schemes adapt fastest to step changes from new large customers? Can one forecast where congestion migrates given announced interconnections? How much error is attributable to unobserved large-load behavior (apply the A1 ceiling method)?	Grid analytics software $\approx$ \$5B $\rightarrow$ $\sim$ \$9B (2030); smart-grid analytics $\approx$ \$10B; US utility capex $\approx$ \$259B (2026), $\sim$ \$1.3T 2026–30; data-center electricity 415 TWh (2024) $\rightarrow$ $\sim$ 945 TWh (2030).	NYISO, PJM, ISO-NE, GE Vernova, Hitachi Energy, Siemens Energy, National Grid, Con Edison, NYPA, Amperon, Yes Energy, Camus Energy, NYSERDA.	PULP/DEC/Fahy energy work; Evans (GE Vernova, forecasting).
A3	Carbon- and grid-aware scheduling of AI/ML workloads ("spinning AI demand")	Treat data-center compute as dispatchable load: learn placement/timing of deferrable AI jobs across sites to follow renewables and relieve congestion.	Data centers run flat; training is deferrable, inference is not; schedulers assume stationary prices; carbon accounting of compute is immature.	What regret can an online scheduler achieve vs. an oracle with perfect LMP and carbon-intensity knowledge? How much stabilization value per flexible MW in NYISO? Is the SAA sample size from your MAT 425 paper sufficient for real-time mixed training/inference dispatch?	Data-center demand doubling to $\sim$ 945 TWh by 2030; hyperscaler capex $>$ \$400B/yr; VPP market $\approx$ \$7B $\rightarrow$ $\sim$ \$17B (2030).	GE Vernova, Google, Microsoft, AWS, NVIDIA, Crusoe, Emerald AI, Verrus, Lancium, ARPA-E, NREL, PNNL.	Evans's published program (direct co-author path); your SAA paper.
A4	Rate-design equity and energy burden with causal ML	Heterogeneous-effect estimation of TOU/CPP rates and DER programs on low-income households; ML-targeted progressive rates (your PULP program, redone with data).	CLCPA/REV affordability mandates; default TOU can raise bills for inflexible households; utilities lack household-level heterogeneity; equity metrics ad hoc.	Which household types are harmed by default TOU, and can nonparametric CATE estimators identify them from AMI + ACS data? What is the welfare-optimal progressive rate under a revenue constraint? How should a \$200M program be targeted to maximize burden reduction per dollar?	Utility customer analytics / demand response (within grid analytics $\approx$ \$5–10B); NY affordability programs in the hundreds of \$M/yr.	NYSERDA, NY DPS/PSC, Uplight, Bidgely, Oracle Utilities (Opower), Arcadia, National Grid, Con Edison, RMI, LBNL, Brattle, E3.	PULP, DEC, Fahy. Evans secondary.
A5	Calibrated LLM pipelines for legislative and regulatory synthesis	Formalize the Fahy-office multi-model pipeline with statistically calibrated error rates (conformal prediction, nonparametric tests) and audit trails.	Hallucinated citations in bill summaries; no error-rate guarantees; unresolved model disagreement; procurement needs evidence.	Can conformal prediction give distribution-free guarantees on factual-error rates of LLM bill summaries? Does multi-model agreement predict correctness, and when does it fail? What human-review sampling policy minimizes cost at a fixed error rate?	GovTech $\approx$ \$880B; RegTech $\approx$ \$20–30B; LLM observability $\approx$ \$2.7B $\rightarrow$ $\sim$ \$9B (2030).	Plural, Quorum, FiscalNote, Palantir, Anthropic and OpenAI public-sector teams, Scale AI, Arize, NY Senate/Assembly, Rockefeller Institute, Code for America.	You built the pipeline; 581/592 theory.
A6	Sequential decisions under uncertainty: poker $\rightarrow$ prediction markets and betting	Online learning/bandits and exploitability analysis applied to prediction-market making, sports pricing and risk limits.	Volume went from $<$ \$5B/mo (Sep 2025) to $\sim$ \$53B/mo (Jul 2026); thin liquidity, manipulation, mispriced long shots; CFTC oversight.	Do no-regret algorithms converge to calibrated prices in thin markets? Can poker-style opponent modeling separate informed from noise traders? What risk-limit rule is optimal under heavy-tailed PnL (formalize your min(500 bb, 3 buy-ins) rule)?	Kalshi + Polymarket $\approx$ \$240B YTD volume (Aug 2026); online betting $\approx$ \$136B; sports betting $\approx$ \$125B; online poker $\approx$ \$4.7B.	Kalshi, Polymarket, Susquehanna (SIG), Jane Street, Jump, Citadel Securities, DraftKings, FanDuel, GTO Wizard, PioSolver, Robinhood.	Poker + econ/game theory. Evans (information theory).
A7	Adaptive K-8 math practice and knowledge tracing	Nonparametric knowledge-tracing to sequence practice for special-ed and enrichment students; measure transfer.	IEP progress monitoring is manual; IXL/DeltaMath are black boxes; equity gaps; teacher time.	Do kernel/nonparametric knowledge-tracing models beat BKT/DKT on small-class data? Can LLM-generated items be validated with IRT at scale?	Adaptive learning $\approx$ \$7.6B; AI tutors $\approx$ \$2.7B; K-12 edtech $\approx$ \$108B.	Khan Academy, IXL, DeltaMath, Desmos/Amplify, Zearn, Carnegie Learning, NWEA, NYSED.	Current teaching role.
A8	Signal-based talent ranking	Learning-to-rank over sparse recruiter "signals" with fairness constraints (your Quantum Talent take-home).	ATS filters miss talent; signal definitions subjective; bias.	Can weak supervision from recruiter rankings learn signal weights that generalize across roles? What is the fairness–accuracy frontier?	AI in HR $\approx$ \$6–8B; AI recruitment $\approx$ \$0.75B.	LinkedIn, Eightfold, HireVue, SeekOut, Greenhouse, Quantum Talent.	Recent project; lower thesis priority.

Table B — Topological Data Analysis × background × market (TDA-led)

#	Topic	Summary	Problems that exist today	Example research questions	Market size (2026, approx.)	Example employers	Fit
B1	Persistent homology of power-grid topology for resilience and layout	Grid as weighted graph/complex filtered by line loading, impedance or voltage: H0 tracks fragmentation, H1 tracks redundant loops; persistence-based resilience scores and layout constraints. Prior art: ORNL network-resilience TDA, NSF grid-robustness PH, MP-Grid (2026) multiparameter PH for outage detection.	N-1/N-k contingency analysis is combinatorial; resilience metrics heuristic; expansion planning ignores redundancy structure; sparse-sensor outage detection.	Does total H1 persistence under a loading filtration predict cascade extent better than betweenness or algebraic connectivity? Can it be a constraint in layout optimization? Do voltage × flow bifiltrations localize outages?	US T&D capex ≈ \$170B (2026); utility capex ≈ \$259B; grid analytics ≈ \$5–10B.	GE Vernova (GridOS), Hitachi Energy, Siemens Energy, NYISO, National Grid, NYPA, EPRI, NREL, PNNL, ORNL, ARPA-E.	Evans (grid layout) + Giunti (multiparameter): natural co-advised topic.
B2	Topology of electricity price and load time series	Sliding-window (Takens) embeddings of LMP/load → VR persistence; H1 detects periodicity and regime structure; bottleneck distance compares zones; vineyards track change over time. "Topological Attention" (NeurIPS 2021) shows viability on electricity demand.	Price spikes and regime shifts; congestion changepoints; zone comparison for basis risk.	Does persistence of delay embeddings flag regime changes (new interconnection, winter storm) earlier than volatility measures? Can bottleneck distance between zonal diagrams define a congestion-similarity metric for FTR/TCC basis risk?	Grid analytics ≈ \$5–10B; FTR/TCC auctions in the billions annually.	NYISO, PJM, DC Energy, Boston Energy Trading, Citadel commodities, Amperon, Yes Energy, Hitachi Energy.	Learning Path Track A; Giunti (stability, vineyards/canopies).
B3	Topological redistricting and voter-file structure	PH and Mapper on district shapes and precinct adjacency complexes (holes = carve-outs); Mapper on canvass/turnout data for field targeting.	Gerrymandering metrics contested; compactness fails on coastlines; campaign segmentation uses k-means.	Does H1 of a district's precinct complex correlate with partisan-intent evidence in court records? Does Mapper surface turnout subgroups that k-means merges (584's cancer-subgroup logic)?	Political/campaign tech ≈ \$1–3B; redistricting litigation.	MGGG Redistricting Lab, Princeton Gerrymandering Project, Catalist, TargetSmart, Civis, L2, Brennan Center, NY Independent Redistricting Commission.	Field director / campaign manager. Giunti.
B4	Coverage holes in urban services (food, retail, community infrastructure)	Čech/alpha complexes on business and service locations (NYC DOHMH inspection pull from your Sept 2026 report) → H0/H1 identify deserts and ring-shaped gaps; Mapper on inspection + review data; vineyards track churn.	Food-desert definitions are grid-based; diaspora information gaps (your G1–G20); siting decisions; language access.	Do persistent H1 features in service-location clouds coincide with measured information/access gaps? How does coverage topology evolve with business churn?	Location intelligence ≈ \$24–30B; foot-traffic analytics ≈ \$8.7B.	Placer.ai, SafeGraph, Esri, Yelp, DoorDash, Uber Eats, NYC DOHMH/DCP, Chinatown BID, Murals for Good partners.	Murals for Good; Asian Food report. Giunti.
B5	Shape of strategy spaces in games (poker solver outputs)	Solver strategies as high-dimensional point clouds over game states; PH/Mapper reveal strategy manifolds; persistence across stack depth / bet sizing is a multiparameter persistence problem.	Solver outputs uninterpretable; human simplifications heuristic; training tools lack structure.	Do equilibria form low-dimensional manifolds with detectable loops (mixed-strategy cycles)? Is persistence across parameters a measure of strategy robustness?	Online poker ≈ \$4.7B; poker-training SaaS ~ \$100M; game-AI research.	GTO Wizard, PioSolver, Run It Once, DeepMind, Meta FAIR, DraftKings.	Poker. Giunti (multiparameter).
B6	Topological anomaly detection in emissions and refrigerant sensor streams	Sublevel-set persistence of sensor series (DEC refrigeration inventory, Rule 494 leak monitoring, smart meters) for noise-robust leak/anomaly detection (1-Lipschitz stability).	High-GWP HFC leaks found late; inventories self-reported; false alarms.	Does persistence-based anomaly scoring cut false positives vs. thresholds or ARIMA residuals? Can Betti curves rank facilities for compliance inspection?	Carbon-accounting software ≈ \$13–28B; climate-risk analytics ≈ \$2–16B.	NYS DEC, EPA, Watershed, Persefoni, Honeywell, Copeland, Carrier, Trane, Daikin, Schneider Electric.	DEC Rule 494. Giunti (SuPerPoV time-series PH).
B7	Persistence-based climate indices → extreme-weather energy demand	Extend SuPerPoV-style polar-vortex scoring to lead-time prediction of cold-snap load spikes and gas-electric stress (Winter Storm Elliott-type events).	Extreme-weather load forecasting; winter reliability in NYISO/ISO-NE; insurers' physical risk.	Does a topological vortex score add lead time to 2-week-ahead peak-load forecasts? Can it price winter reliability products?	Climate-risk analytics ≈ \$2–16B; grid analytics; reinsurance.	NOAA, NCAR, NYISO, ISO-NE, Swiss Re, Munich Re, Jupiter Intelligence, First Street, Amperon.	Giunti directly; energy policy.

Table C — All four together: ML + TDA + background + market

#	Topic	Summary	Problems that exist today	Example research questions	Market size (2026, approx.)	Example employers	Fit
C1	Topology-aware load and congestion forecasting under data-center-driven shift ★	PH features from (a) the grid network under a loading filtration and (b) delay embeddings of nodal prices/loads, fed to boosted/online forecasters (581/592); evaluate under distribution shift from large-load interconnections; use the stability theorem (584) as the robustness argument.	Load growth breaks forecasts; congestion migrates as topology changes; features are ad hoc; ISOs lack interpretable shift detectors.	Do topological features improve day-ahead congestion forecasts vs. GBM baselines (MASE, pinball)? Does Lipschitz stability of PH yield measurably smaller degradation under shift than raw features? Can vineyards/canopies of zonal diagrams flag regime changes before price statistics do?	Grid analytics ≈ \$5–10B; US utility capex ≈ \$259B (2026), ~\$1.3T 2026–30; data-center electricity → ~945 TWh (2030).	GE Vernova, NYISO, PJM, Hitachi Energy, Siemens Energy, Amperon, Camus, National Grid, Con Edison, Google/Microsoft energy teams, NREL, PNNL.	Strongest overall: both professors' current work, your energy-policy history, all four courses. Data: NYISO OASIS, EIA-930, PJM Data Miner.
C2	Topological regime detection for carbon-aware AI dispatch across distributed data centers	Build on Evans's "Spinning AI Demand": SAA scenario clouds (prices, renewables, load) → PH detects multimodality/regime shifts that change the optimal schedule; online learning for job placement; price × carbon-intensity bifiltration.	Flat data-center load; schedulers assume stationarity; SAA sample size (your MAT 425 paper) chosen heuristically; noisy carbon signals.	Can PH of scenario clouds certify SAA sufficiency (persistent clusters stabilize ⇒ N large enough)? Does topological regime detection trigger re-optimization with lower regret than fixed intervals? What stabilization value per flexible MW in NYISO?	Data-center demand → ~945 TWh (2030); hyperscaler capex > \$400B/yr; VPP ≈ \$7B → ~\$17B (2030).	GE Vernova, Google, Microsoft, AWS, NVIDIA, Crusoe, Emerald AI, Verrus, Lancium, ARPA-E, NREL.	Evans's exact program; your SAA paper; Giunti for the topology.
C3	Topological structure of marketplace delivery networks	Seller→customer route graph of Olist (then a US dataset) filtered by volume/transit time; H1 loops and Mapper on orders locate structurally slow sub-networks; canopies/vineyards track topology over time (the May 2018 truckers' strike as a topological event); topological features tested against your R ² ceiling.	6-day unexplained residual; promises as liability buffers; unobserved network changes; opaque carrier sub-network performance.	Does topological position (persistent H1 cycle / Mapper node membership) explain variance beyond the route+week ceiling? Do vineyards detect the strike and Carnival shocks? Do PH features transfer across marketplaces?	Last-mile ≈ \$200B; supply-chain analytics ≈ \$14B; last-mile software ≈ \$3B.	Amazon, Walmart, Shopify, Mercado Libre, Olist, Flexport, project44, FourKites, UPS, FedEx.	Lowest startup cost: data and code exist. Giunti (canopies).
C4	PH-enhanced evaluation of LLM policy pipelines	Vectorize embedding clouds of bills, comments and model outputs with PH (persistence images/landscapes) as features for hallucination prediction and corpus coverage-gap detection; wrap with conformal calibration (581). Extends Giunti–Raghavaraju–Rieck (2026).	Hallucination detection heuristic; corpus coverage unknown; evaluation sets small/private; legislative use requires audit trails.	Do topological features of output-embedding clouds predict factual-error rates better than embedding norms or entropy? Do persistent H1 features in a regulatory corpus mark unaddressed topics? Can conformal prediction wrap the detector with guarantees?	LLM observability ≈ \$2.7B → ~\$9.3B (2030); GovTech ≈ \$880B; RegTech ≈ \$20–30B.	Anthropic, OpenAI, Scale AI, Arize, Galileo, Weights & Biases, FiscalNote, Plural, Palantir, Rockefeller Institute, state legislatures.	Giunti's June 2026 preprint; your pipeline; 581/592 theory. High upside.
C5	Topological decision analytics for prediction markets and sequential games	PH/Mapper on order-book and price-path embeddings and on solver strategy spaces; online learning for adaptive pricing; manipulation detection via topological changepoints.	Thin-market manipulation; event-driven regime shifts; uninterpretable strategies; CFTC oversight of ~\$50B/month markets.	Does PH of price-path embeddings detect manipulation/insider regimes better than volatility and volume anomalies? Do no-regret market makers using topological regime features suffer less adverse selection? Is strategy-space persistence across stack sizes a robustness metric?	Kalshi + Polymarket ≈ \$240B YTD (Aug 2026); online betting ≈ \$136B; online poker ≈ \$4.7B.	Kalshi, Polymarket, SIG, Jane Street, Citadel Securities, DraftKings, FanDuel, GTO Wizard, CFTC.	Poker + econ; Giunti (multiparameter); Evans (information theory).
C6	Energy-burden archetypes: Mapper + causal ML on household energy data	Mapper on AMI load shapes × ACS income/housing to find household archetypes (584's subgroup logic); heterogeneous TOU/DER effects per archetype; income × usage bifiltration for stable archetypes.	TOU harms some low-income households; k-means hides subgroups; CLCPA affordability mandates; program targeting.	Does Mapper reveal subgroups with opposite-sign treatment effects that k-means merges? Does archetype-conditioned CATE improve burden reduction per dollar of a progressive rate program? Are archetypes stable (bottleneck distance) across utilities?	Grid/customer analytics ≈ \$5–10B; NY affordability programs; demand-response market.	NYSERDA, NY DPS, Uplight, Biddely, Oracle Utilities, Arcadia, National Grid, Con Edison, LBNL, RMI, Brattle.	PULP \$200M program, DEC, Fahy; Giunti (Mapper) + Evans (grid). Data: Pecan Street, London Low Carbon, CER Ireland.
C7	Grid-layout optimization with topological redundancy constraints ★	Persistence-based regularizers (total H1 persistence under a loading filtration) inside expansion/layout optimization; ML surrogate for OPF (Evans's sparse-tableau OPF) for tractability; validate on IEEE test systems and synthetic NY grids.	Expansion planning is MILP-heavy and ignores redundancy structure; resilience vs. cost; cascading failures.	Does a persistence regularizer yield layouts with better N-k survivability per dollar than connectivity-based regularizers? Can an ML OPF surrogate plus topological constraint solve at planning scale? Do voltage × flow multiparameter summaries (MP-Grid 2026) localize fragility?	US T&D capex ≈ \$170B (2026); utility capex ~\$1.3T 2026–30; grid analytics ≈ \$5–10B.	GE Vernova, Hitachi Energy, Siemens Energy, NYISO, NYPA, National Grid, EPRI, NREL, PNNL, ARPA-E, Quanta, Burns & McDonnell.	Evans's stated focus (grid layout) + Giunti's multiparameter work + your PSC/energy background.
C8	Knowledge-space topology for adaptive math learning	Mapper/PH on item-response matrices to map knowledge states; nonparametric knowledge tracing to sequence practice.	Progress monitoring manual; adaptive tools opaque; equity gaps.	Do topological knowledge maps predict transfer better than skill taxonomies? Does Mapper reveal misconception clusters?	Adaptive learning ≈ \$7.6B; AI tutors ≈ \$2.7B.	Khan Academy, IXL, DeltaMath, Desmos/Amplify, Carnegie Learning, NWEA.	Teaching role; Giunti. Lower market pull.

Recommendation and sequencing

1. C1 (topology-aware grid forecasting) is the best single bet: it is co-advisable (Evans for grid and forecasting, Giunti for vineyards/stability), uses every course, sits in a market with the strongest hiring tailwind (utility capex, data-center load), and public data exists (NYISO, PJM, EIA-930).

2. C7 (layout optimization with topological constraints) is the most direct match to Evans's stated focus and to Giunti's multiparameter work, with a fresh literature base (MP-Grid 2026). Higher technical risk; strong thesis-level novelty.

3. C3 (delivery-network topology) is the cheapest to start because your 592 code and the Olist data already exist, and it gives Giunti a concrete canopies/vineyards application. Good as a first paper or a warm-up for C1.

4. C4 (PH-enhanced LLM evaluation) is the high-upside wildcard: it extends Giunti's newest preprint, reuses your legislative LLM pipeline, and targets the fastest-growing software market in the list.

A practical path: start C3 in Fall 2026 as the "TDA + ML on data you already own" warm-up, pitch C1/C7 to both professors as the S27 project, and keep C4 as a side paper.

Market-size and background sources

- Grid analytics: [Yahoo Finance / TBRC grid analytics software \\$5.08B \(2026\)](#); [smart-grid data analytics → \\$9.7B by 2030](#); [Research and Markets smart grid analytics](#).
- Utility capex and T&D: [S&P Global: ~\\$1.3T US utility capex 2026-30](#); [Deluair: >\\$170B T&D in 2026](#).
- Data-center electricity: [S&P Global on IEA: 415 TWh \(2024\) → 945 TWh \(2030\)](#).
- VPP: [Grand View Research VPP market](#); [Mordor Intelligence VPP](#).
- Last-mile and supply-chain analytics: [Grand View last-mile delivery](#); [Meticulous supply-chain analytics \\$14.18B \(2026\)](#); [ClickPost last-mile statistics](#).
- GovTech / RegTech: [TBRC GovTech \\$882B \(2026\)](#); [Research and Markets RegTech](#); [Future Market Insights RegTech](#).
- LLM observability: [Research and Markets LLM observability \\$2.69B \(2026\)](#); [openPR 36.2% CAGR to 2030](#).
- Prediction markets and betting: [Pew Research on Kalshi/Polymarket volume](#); [CNBC World Cup volumes](#); [TBRC sports betting](#); [Grand View online poker](#).
- Education: [Fact.MR adaptive learning \\$7.6B \(2026\)](#); [Grand View AI tutors](#); [K-12 edtech \\$108B](#).
- HR tech: [Grand View AI in HR](#); [Pin AI recruitment market](#).
- Location intelligence: [Fortune Business Insights](#); [Straits Research location analytics](#); [foot-traffic analytics \\$8.7B](#).
- Carbon accounting and climate risk: [Grand View carbon accounting software](#); [Fortune Business Insights climate risk analytics](#); [GM Insights climate risk management](#).
- Prior art for grid TDA: [MP-Grid: detecting power grid outages with topological machine learning \(2026\)](#); [Role of local geometry in robustness of power grid networks](#); [ORNL TDA for network resilience](#); [Topological Attention for Time Series Forecasting \(NeurIPS 2021\)](#).
- Professors: [Scott Evans, UAlbany Math](#); [Sustainable Grid through Distributed Data Centers \(arXiv 2504.03663\)](#); [Power-grid-aware ML device patent](#); [Sparse tableau OPE](#); [Barbara Giunti CV](#); [Canopies \(arXiv 2606.23859\)](#); [Comparing Chatbot Performance Enhanced with Persistent Homology \(arXiv 2606.29857\)](#); [Fast queries of fibered barcodes](#); [Efficient two-parameter persistence via cohomology](#).
- Courses: [UAlbany graduate bulletin, Mathematics and Statistics](#); [AMAT 584 TDA II course page](#); [Data Science MS program](#).